

Test Plan Document

Smart Expense & Budget Management System

1. Purpose

This document lists the test cases written for the implemented use cases of the Smart Expense & Budget Management System. It covers unit, integration and system test cases. Each case is run manually and the Actual Result and Test Result columns are filled in after the run.

2. Test Environment

Application	Smart Expense & Budget Management System
Testing type	Manual testing
Browsers	Chrome and Firefox, latest versions, on desktop and mobile
Test accounts	Two separate accounts, used to check that one user cannot see another user's data
Test data	Sample values only. No real financial data is used in testing.

3. Column Meanings

Test Case ID	Unique ID for each test case. UT = unit test, IT = integration test, ST = system test.
Name of the module	The main module or sub module being tested.
Test Case Description	Short summary of what the test is checking.
Pre-Conditions	Anything that must be done or must exist before the test can run.
Test Steps	The steps to follow, in the order they are to be carried out.
Test Data	The input used for the test. Both valid and invalid values are covered across the cases.
Expected Results	What should happen, including the exact message shown on screen.
Actual Result	Filled in after the test is run.
Test Result	Pass if the actual result matches the expected result, otherwise Fail.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_01	User Registration & Login	Test registration with valid details	App is open in Chrome on the registration page	1: Open the sign up page 2: Enter name, email and password 3: Click Register	Name: Test User Email: user1@test.com Password: Test@1234	Account is created and the user is taken to the dashboard		

4. Test Cases

1. User Registration & Login (UC-01)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_02	User Registration & Login	Test login with a valid account	A registered account already exists	1: Open the login page 2: Enter email and password 3: Click Login	Email: user1@test.com Password: Test@1234	Login succeeds and the dashboard is displayed		
UT_03	User Registration & Login	Test login with a wrong password	A registered account already exists	1: Open the login page 2: Enter correct email with a wrong password 3: Click Login	Email: user1@test.com Password: Wrong@999	Login is refused and the message "Invalid email or password" is shown		
UT_04	User Registration & Login	Test registration with a duplicate email	user1@test.com is already registered	1: Open the sign up page 2: Enter the same email again 3: Click Register	Email: user1@test.com	Registration is refused with the message "Email already registered"		
UT_05	User Registration & Login	Test that required fields are validated	App is on the registration page	1: Leave the email and password blank 2: Click Register	Email: (blank) Password: (blank)	Form is not submitted and both fields are marked as required		
UT_06	User Registration & Login	Check that the password is not stored in plain text (SR02)	One account has been registered	1: Register a new account 2: Open the users table in the database 3: Look at the password column	Password: Test@1234	The stored value is a hash, the original password is not readable		
IT_01	Login + Dashboard	Check that the dashboard loads the logged-in user's own data	Two accounts exist, each with its own expenses	1: Log in as user1 2: Note the totals on the dashboard 3: Log out and log in as user2	user1@test.com, user2@test.com	Each user sees only their own totals, no data from the other account		
IT_02	Login + Session	Check that an idle session expires (SR07)	User is logged in and the timeout is set	1: Log in 2: Leave the app idle past the timeout 3: Click any menu item	Idle time: just over the set timeout	The session ends and the user is sent back to the login page		
ST_01	End to end	Check that a user cannot reach data pages without logging in (SR01)	No active session in the browser	1: Clear cookies 2: Paste the dashboard URL directly in the address bar	URL: /dashboard	The app redirects to login and shows no financial data		

2. Income & Expense Records (UC-02, UC-03)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_07	Income & Expense Records	Test adding a valid expense	User is logged in and at least one category exists	1: Click Add Expense 2: Enter amount, date and category 3: Click Save	Amount: 450 Date: today Category: Food	Expense is saved and appears at the top of the expense list		

UT_08	Income & Expense Records	Test adding an expense with a negative amount	User is on the Add Expense form	1: Enter a negative amount 2: Click Save	Amount: -200	Entry is refused with the message "Amount must be greater than zero"		
UT_09	Income & Expense Records	Test adding an expense with a future date	User is on the Add Expense form	1: Pick a date after today 2: Click Save	Date: today + 10 days	Entry is refused with the message "Date cannot be in the future"		
UT_10	Income & Expense Records	Test editing an existing expense	At least one expense already exists	1: Open the expense list 2: Click Edit on an entry 3: Change the amount and save	Old amount: 450 New amount: 600	The updated amount is shown in the list and the totals change accordingly		
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_11	Income & Expense Records	Test deleting an expense	At least one expense already exists	1: Open the expense list 2: Click Delete 3: Confirm	Expense: 600, Food	Entry is removed from the list and no longer counted in the totals		
UT_12	Income & Expense Records	Test adding an income record	User is logged in	1: Click Add Income 2: Enter amount and source 3: Click Save	Amount: 15000 Source: Part-time job	Income is saved and the available balance goes up by the same amount		
UT_13	Income & Expense Records	Test the category suggestion (FR05)	The suggestion feature is switched on	1: Open Add Expense 2: Type a merchant name in the note field	Note: Metro Station	A category such as Travel is suggested, and the user can still change it		
IT_03	Expense + Budget	Check that adding an expense updates the remaining budget	A monthly budget of 10000 is set with 2000 already spent	1: Note the remaining budget 2: Add an expense of 500 3: Look at the remaining budget again	Expense: 500	Remaining budget drops from 8000 to 7500 without needing a page refresh		
IT_04	Expense + Charts	Check that a new expense shows up in the charts	The dashboard charts are already loaded	1: Note the Food slice in the pie chart 2: Add an expense under Food 3: Reload the dashboard	Expense: 300, Food	The Food slice grows and the chart total matches the expense list		
ST_02	End to end	Check a user cannot open another user's expense (SR04)	Two accounts exist, each with expenses	1: Log in as user1 2: Note an expense ID belonging to user2 3: Request that ID directly	URL: /expense/{user2 id}	The request is refused and no data from the other account is returned		

3. Budget Management & Alerts (UC-04)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_14	Budget Management	Test setting a monthly budget	User is logged in	1: Open Budgets 2: Enter a monthly limit 3: Click Save	Monthly budget: 10000	The budget is saved and shown on the dashboard for the current month		

UT_15	Budget Management	Test setting a zero or negative budget	User is on the Budgets page	1: Enter 0 as the budget 2: Click Save	Budget: 0	The value is refused with a message asking for an amount above zero		
UT_16	Budget Management	Test setting a category-wise budget	A monthly budget already exists	1: Open Budgets 2: Set a limit for the Food category 3: Save	Food budget: 3000	The Food limit is saved and tracked separately from the monthly budget		
UT_17	Budget Management	Test the remaining budget calculation (FR08)	Budget 10000 with expenses of 2500 recorded	1: Open the dashboard 2: Read the remaining budget figure	Budget: 10000 Spent: 2500	Remaining budget shows 7500		
IT_05	Budget + Alerts	Check that the warning alert fires at the threshold (FR09)	Budget 10000 with 7500 already spent and the warning set at 80%	1: Add an expense of 600 2: Watch for an alert	Expense: 600 (takes the total to 8100)	A warning alert is shown saying the user is close to the budget limit		
IT_06	Budget + Alerts	Check the alert when the budget is crossed	Budget 10000 with 9800 already spent	1: Add an expense of 500 2: Watch for an alert	Expense: 500	An over-budget alert is shown and the remaining budget is displayed as -300		
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
IT_07	Budget + Notification	Check that only one alert is sent per threshold crossing	Budget already crossed once and an alert shown	1: Add two more small expenses 2: Count the alerts received	Expenses: 50, 80	No duplicate alerts are sent for the same threshold		
ST_03	End to end	Full budget cycle from setting a limit to being alerted	Fresh account with no data	1: Set a monthly budget 2: Add expenses until the limit is crossed 3: Check the dashboard and alerts	Budget: 5000 Expenses: 2000, 2000, 1500	Remaining budget updates each time and an over-budget alert appears on the last entry		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
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UT_24	Savings Goals	Test creating a savings goal (FR14)	User is logged in	1: Open Savings Goals 2: Enter a name and target amount 3: Save	Goal: New laptop Target: 50000	The goal is created and shows 0% progress		
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4. Analytics, Prediction & Anomaly Detection (UC-05)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_18	Analytics & Charts	Test the pie chart totals by category (FR10)	Expenses exist across at least three categories	1: Open the dashboard 2: Compare each pie slice with the expense list	Food 1200, Travel 800, Bills 2000	Each slice matches the category total in the expense list		
UT_19	Analytics & Charts	Test the monthly trend line chart	Expenses exist for at least three months	1: Open Analytics 2: Select the last 3 months	Period: last 3 months	A line chart is drawn with one point per month and correct monthly totals		
UT_20	Anomaly Detection	Test that an unusual expense is flagged (FR11)	The user normally spends 200 to 600 on Food	1: Add an unusually large Food expense 2: Look for a flag on the entry	Amount: 9000, Food	The entry is flagged for review with a note asking the user to confirm the amount		
UT_21	Anomaly Detection	Test that a normal expense is not flagged	Same history as UT_20	1: Add a typical Food expense 2: Look for a flag	Amount: 350, Food	The entry is saved normally with no flag		
UT_22	Prediction	Test the monthly expenditure forecast (FR12)	At least three months of expense history exist	1: Open the dashboard 2: Read the forecast for the current month	History: 8000, 8500, 9000	A forecast is shown in a sensible range with the months it is based on		
UT_23	Prediction	Test the forecast for a new account with no history	A fresh account with under one month of data	1: Open the dashboard 2: Look at the forecast area	History: 4 days of data	No forecast is shown, instead a message saying more history is needed		
IT_08	Prediction + Recommendation	Check the recommended budget stays within income (FR13)	Income and at least three months of expenses recorded	1: Open Budgets 2: Click Recommend Budget 3: Compare the total with the income	Income: 15000 Average spend: 9000	A category-wise suggestion is shown and the total does not exceed the income		
ST_04	End to end	Check that the health score reacts to a change in habits (FR18)	An account with a few months of steady spending	1: Note the current score 2: Add a month of heavy overspending 3: Check the score again	Extra expenses: 3 x 5000	The score drops and a short reason for the change is shown		

5. Savings Goals & Recurring Expenses (UC-06, UC-07)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_25	Savings Goals	Test the time-to-goal estimate (FR15)	A goal exists and the user has a steady saving rate	1: Open the goal 2: Read the estimated time	Target: 50000 Saving rate: 5000 a month	An estimate of about 10 months is shown		

UT_26	Savings Goals	Test the estimate when nothing is being saved	A goal exists but the user saved nothing last month	1: Open the goal 2: Read the estimated time	Saving rate: 0	No number is shown, instead a message that the goal cannot be reached at the current rate		
UT_27	Recurring Expenses	Test creating a recurring expense (FR16)	User is logged in	1: Open Recurring 2: Enter amount, category and cycle 3: Save	Amount: 6000 Category: Rent Cycle: monthly	The recurring expense is saved and listed with its next due date		
UT_28	Recurring Expenses	Test that a recurring expense is not created twice in one cycle	A monthly recurring expense already exists	1: Move the system date into the next cycle 2: Open the expense list	Cycle: monthly	Exactly one entry is created for the new cycle, not two		
IT_09	Recurring + Reminder	Check the reminder before a due date (FR17)	A recurring expense is due in a few days	1: Move the date to within the reminder window 2: Open the app	Due in: 3 days	A reminder is shown for the upcoming payment		
IT_10	Recurring + Reminder	Check that reminders stop once the payment is marked paid	A reminder is currently showing	1: Mark the payment as paid 2: Reopen the app	Status: paid	The reminder no longer appears for that cycle		

6. Simulator, Reports & Cross-cutting Checks (UC-08, UC-09, UC-10)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_29	What-If Simulator	Test simulating an affordable purchase (FR19)	Budget and expense data exist for the month	1: Open the simulator 2: Enter a planned purchase 3: Run	Purchase: 2000 Remaining budget: 7500	The result shows the budget left after the purchase and marks it as affordable		
UT_30	What-If Simulator	Test simulating a purchase the user cannot afford	Same starting data as UT_29	1: Open the simulator 2: Enter a large purchase 3: Run	Purchase: 40000	The result warns that the purchase goes past the budget and shows the shortfall		
UT_31	What-If Simulator	Check that a simulation is not saved as a real expense	A simulation has just been run	1: Run a simulation 2: Open the expense list	Purchase: 40000	No new expense appears in the list and the remaining budget is unchanged		
UT_32	Export Reports	Test exporting a report (FR20)	The account has at least one month of data	1: Open Reports 2: Pick a month 3: Click Export	Month: last month Format: PDF	A file downloads with figures matching the dashboard for that month		
UT_33	Export Reports	Test exporting a month with no data	A month with no expenses is selected	1: Open Reports 2: Pick an empty month 3: Click Export	Month: with no entries	The app shows a message that there is nothing to export instead of an empty file		
IT_11	Input Validation	Check that unsafe input is rejected (SR05)	User is on the Add Expense form	1: Enter a script tag in the note field 2: Save 3: Reopen the entry	Note: <script>alert(1)</script>	The input is rejected or escaped, and nothing is executed when the entry is opened		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
ST_05	End to end	Full user journey from registration to export	Fresh account	1: Register and log in 2: Add income and a few expenses 3: Set a budget 4: Check the dashboard 5: Export the report	Income: 15000 Expenses: 500, 1200, 800 Budget: 10000	Every step works in order and the exported figures match what the dashboard showed		
ST_06	End to end	Check that records survive a restart (NFR10)	Several expenses have been saved	1: Note the totals 2: Restart the server 3: Log in and check again	Expenses: 5 entries	All entries are still present and the totals are unchanged		

5. Summary

Type of Testing	Number of Test Cases	Covers
Unit Testing (UT)	33	Individual functions and form validations
Integration Testing (IT)	11	Modules working together, e.g. expense entry updating the budget
System Testing (ST)	6	Complete user journeys and cross-cutting behaviour
Total	50	

Note: the Actual Result and Test Result columns are left blank on purpose. They are to be filled in by hand while running each test case, as required by the instructions.